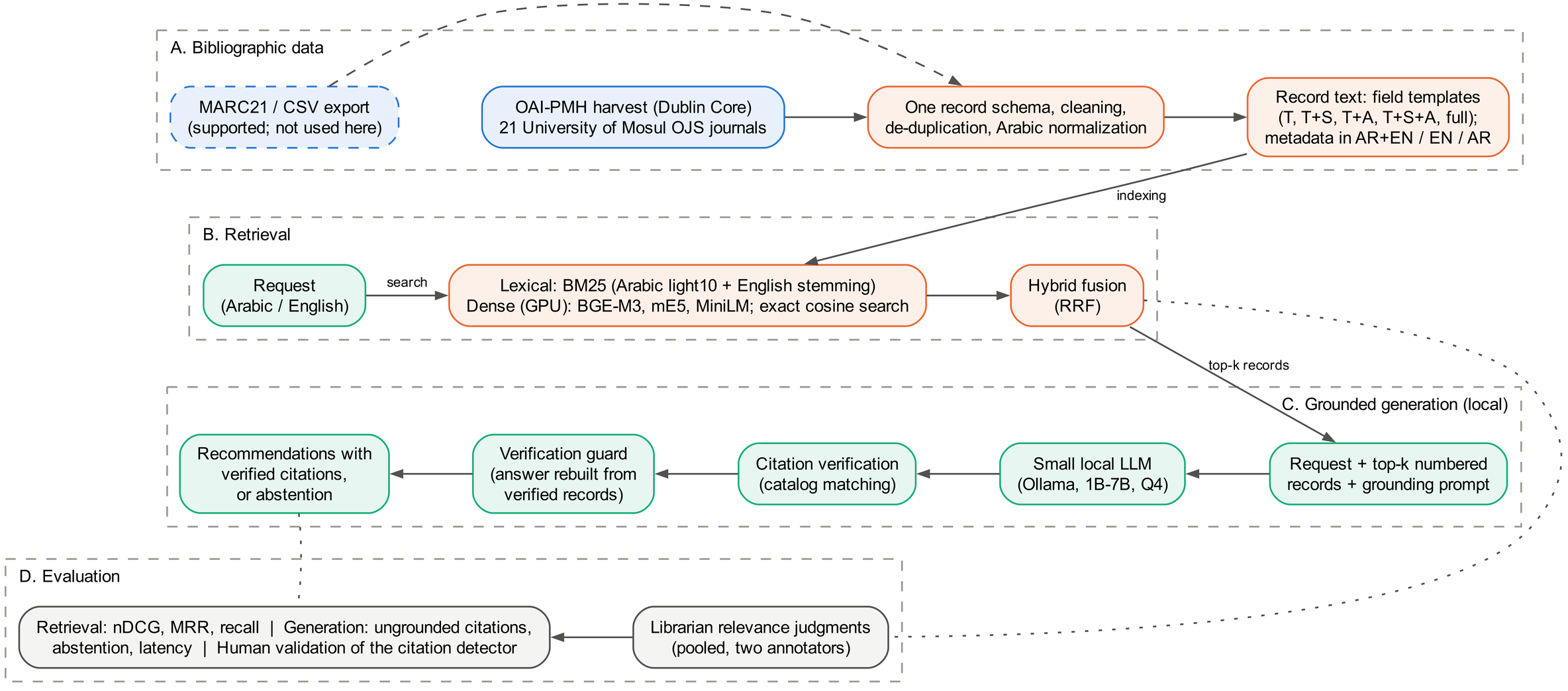
A. Bibliographic data
MARC21 / CSV export
(supported; not used here)
OAI-PMH harvest (Dublin Core)
21 University of Mosul OJS journals
One record schema, cleaning,
de-duplication, Arabic normalization
Record text: field templates
(T, T+S, T+A, T+S+A, full);
metadata in AR+EN / EN / AR
indexing
B. Retrieval
Request
(Arabic / English)
search
Lexical: BM25 (Arabic light10 + English stemming)
Dense (GPU): BGE-M3, mE5, MiniLM; exact cosine search
Hybrid fusion
(RRF)
top-k records
C. Grounded generation (local)
Request + top-k numbered
records + grounding prompt
Small local LLM
(Ollama, 1B-7B, Q4)
Citation verification
(catalog matching)
Verification guard
(answer rebuilt from
verified records)
Recommendations with
verified citations,
or abstention
D. Evaluation
Librarian relevance judgments
(pooled, two annotators)
Retrieval: nDCG, MRR, recall | Generation: ungrounded citations,
abstention, latency | Human validation of the citation detector